Algebra: Diagnostic A

Time allowed: 45 minutes **Total:** 14 diagnostic marks

Instructions Answer both questions. Show full working and give complete proofs. No calculator, internet, AI assistance or hints during an independent diagnostic. Use separate answer paper if required.

Candidate: _____ Date: _____

Question 1 [7 marks]

ALG-01

Let $f : \mathbb{Z} \to \mathbb{Z}$ satisfy

$$f(x + y) + f(x - y) = 2f(x) + 2f(y)$$

for every pair of integers x and y.

Determine every such function f, and prove that your list is complete.

Question 2 [7 marks]

ALG-02

Let a, b, c be positive real numbers. Prove that

$$(a + b + c)^3 + 9abc \geq 4(a + b + c)(ab + bc + ca).$$

Determine all cases of equality.

End of paper. Write your answers on separate paper, labelled with the question number.